Jal dristhi

Feasibility Report on Rooftop Rainwater Harvesting Systems

📍 Site Overview

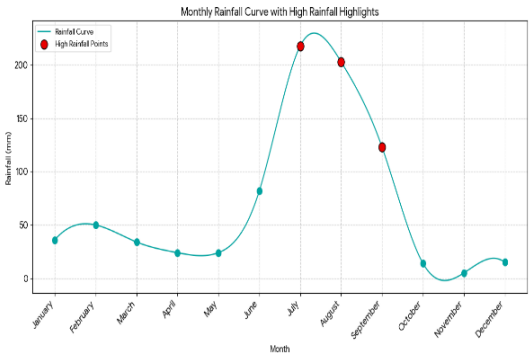
1. Location: Shanti Villa, 42/A Rosewood Lane, Sector 14, Karnal, Haryana – 132001
2. Roof Area: 110 m²(approx. 119.6 GAJ)
3. Family Size: 5 members
4. Rooftop Type: RCC flat roof with parapet

Output

1.Rainfall Data

✅ **Feasibility Rating: GOOD**

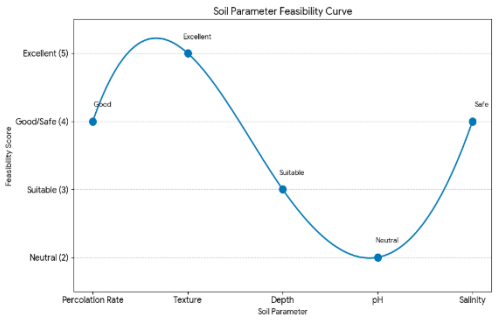
Month	Rainfall(mm)
January	36.00
February	50.00
March	34.00
April	24.00
May	24.00
June	82.00
July	218.00
August	203.00
September	123.00
October	14.00
November	5.00
December	15.00



2. Soil Feasibility

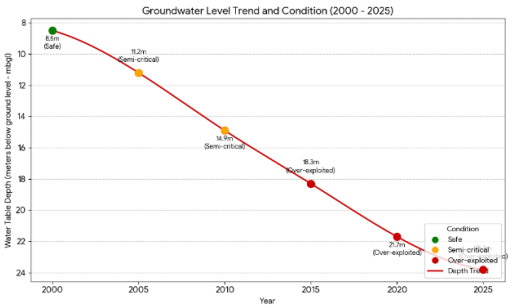
✅ **Feasibility Rating: GOOD**

Parameter	Ideal Range	Feasibility
Percolation Rate	10–30 mm/hour	✅ Good
Texture	Loam / Sandy Loam	✅ Excellent
Depth	>2 meters	✅ Suitable
pH	6.5–8.5	✅ Neutral
Salinity	EC < 2 dS/m	✅ Safe



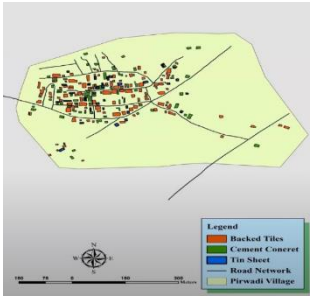
📊 Groundwater Level Data

Year	Water Table Depth(mbgl)	Condition
2000	8.5m	Safe
2005	11.2m	Semi-critical
2010	14.9m	Semi-critical
2015	18.3m	Over-exploited
2020	21.7m	Over-exploited
2025	23.8m	Over-exploited

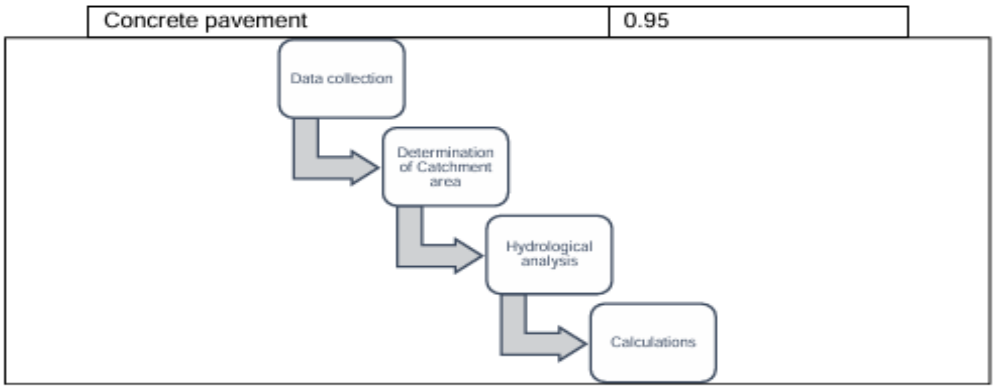


3.Runoff Coefficient

Runoff Coefficient for your rooftop= 0.85



Implementation



Collecting Rainwater from Your Roof

Rainwater Harvested=Area× Rainfall× Runoff Coefficient

= 100 m2×0.828 m×0.85

= ≈ 70,380 Liters.

Recharge Pit Design

Types of Recharge Pit Design we can use here:-

1.Percolation Pit
2. Recharge Shaft
3.Soak Well
4. Recharge Trench
5. Injection Well

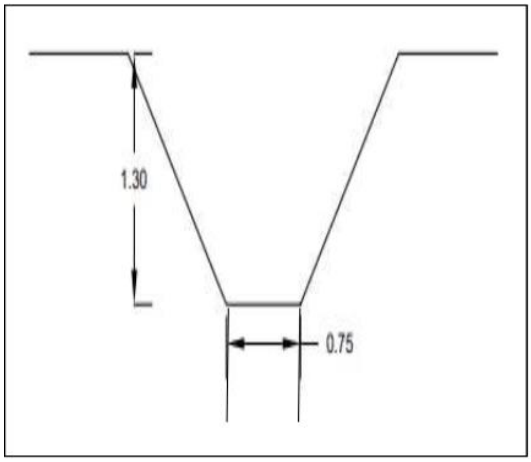
Best pit suitable for this location:- Recharge Shaft

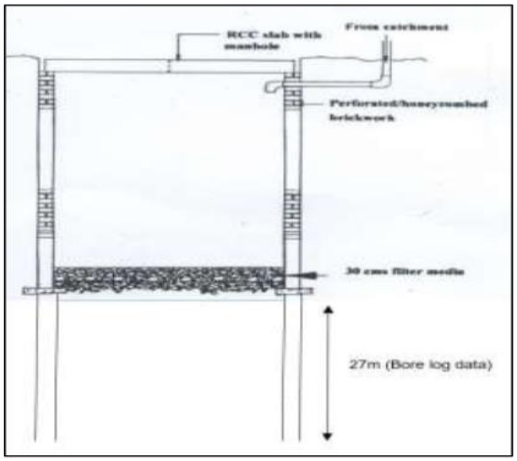
How to build Recharge pit design:-

1. Site Selection: Choose a low-lying area with permeable soil to allow water infiltration.

2.Determine Pit Size**:** Use rainfall data and catchment size to decide dimensions.

Example: For moderate rainfall and a small rooftop, a 1.5m × 1.5m × 2m pit is sufficient excavation .





3.Dig the pit to the required dimensions: Length 1.5m, Breadth 1.5m, Depth 2m.

4.Wall Lining (Optional)

Line the walls with brick or stone masonry.

Include weep holes to allow water to seep through.

5.Filter Media Filling

Fill the pit in layers:

Bottom: Brick bats or boulders (0.5m)

Middle: Charcoal or gravel (0.5m)

Top: Coarse sand (1.0m)

6.Inlet Pipe & Slit Trap

Connect a rainwater inlet pipe from the catchment area.

Install a slit trap to remove debris before water enters the pit.

7. Covering: Place a concrete slab or metal grill over the pit for safety and access.

8. Maintenance

Clean the slit trap regularly.

Replace filter media every 1–2 years or as needed.

Tank Size

1. **Underground RCC tank:** 5,000-liter capacity.
2. **Overflow outlet:** Connected to recharge pit.

Cost Analysis & Subsidy

Component	Specification	Estimated Cost(INR)
Excavation & Earthwork	Manual digging of 10-15 m deep shaft	6,000- 8,000
Casing Material	RCC rings or PVC pipe (1 m diameter)	10,000 - 12,000
Filter Media	Gravel , coarse sand , charcoal	3,000- 4,000
First Flush Diverter	PVC valve system to discard initial runoff	1,500 - 2,000
Collection Pipes	6-inch PVC pipes (20-30 m length)	3,000 - 4,000
Storage Tank (Optional)	5,000- liter RCC tank for reuse	18,000 - 22,000
Labor & Installation	Skilled + unskilled labor	6,000 - 8,000
Miscellaneous	Sealants, Cover ,fittings	2,000 - 3,000
Total Estimated Cost	Without tank : 31,500 - 41,000	
	With tank : 49,500 - 63,000	

Subsidy Eligibility: Up to Rs. 20,000

Contact Contractor: 9996756870